

Chen Xiaoyu

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Agent Engineer | Generalist Agents | Investment-Research Agents

Backend / Agent engineer focused on bringing **runnable, observable, evaluable, and iterable** Agent systems to production and to complex decision domains.

At Alibaba Cloud, led core pipeline development for a real-time voice Agent, cutting **end-to-end dialogue latency from 3s+ to 1.5–1.8s** while stably serving 100K+ daily calls in a high-concurrency production environment.

Outside of work, four years of independent investment-research Agent work — **Uteki** studies “well-covered” high-quality companies, **Shinkai** discovers “under-covered” opportunities; both are built around **evidence constraint, temporal consistency, structured reasoning, execution trace, and multi-model evaluation**, forming a Research → Discovery closed loop.

EDUCATION

Xi'an Jiaotong University · Software Engineering M.S. 2020.09 – 2023.07 (School of Telecommunications) · B.S. 2016.09 – 2020.06 (School of Software)

WORK EXPERIENCE

Alibaba Cloud Intelligence · Backend / Agent Engineer 2023.07 – Present, pivoted to Agent in 2024.10

AGENT PROJECTS

Marketing Voice Agent · Alibaba Cloud Digital Marketing Human

Project: aliyun.com/product/thirdsw/aiemployee

Real-time voice Agent for enterprise automated outbound calls, sales conversion, and 7×24 customer interaction. As lead backend developer, drove pipeline construction and iteration of the outbound-call system, focused on **low latency, interruptibility, multi-turn consistency, and platform extensibility**.

Inference pipeline & latency

- End-to-end streaming architecture: **ASR → LLM decision (tool calling + RAG) → TTS**, reducing wait time and first-packet latency
- vLLM acceleration + LLM Cache, **end-to-end latency 3s+ → 1.5–1.8s**
- Dual VAD + dual-path ASR + denoising model (Omni3), systematic handling of barge-in, proper-noun recognition, and noisy environments

Platform & strategy iteration

- Upgraded a single-purpose outbound flow into a **configurable Agent platform**: compose system prompt + knowledge base + tools + execution policy per business goal; function calling + RAG framework with auto-generated QA pairs from corpora
- Online debugging + version management + A/B canary, **strategy iteration cycle shortened from days to hours**

Scale: 100K+ daily outbound calls · **200 QPM peak** (~10K/hr) · **200+ stable concurrent calls**

Investment-Research Agent · Uteki

Project: v1 | v2 | v3

Investment-research decision infrastructure for **S&P 500 / Nasdaq 100** high-quality companies — audit-ready, replayable, quantitatively evaluable; produces structured dossiers and decision-evidence chains as trustworthy ground truth for strategy validation.

Core design principles

- Evidence before conclusion** — Agent does not output verdicts directly; it first builds a SourceCatalog unifying financials, market data, news, and search results into DataPoints carrying *source, timestamp, confidence*
- Staged reasoning** — decompose company research into a 7-Gate decision chain: **Business Quality → Fisher Growth → Buffett Moat → Munger Management → Inversion Test → Valuation → Synthesis**
- Constrained autonomy** — each gate can call tools, but output must contain structured conclusion + confidence + `[src:N]` citations; parser validates citation legitimacy, blocking unsourced reasoning
- Temporal consistency** — `as_of` time-window constrains data and source timestamps for honest historical backtesting, **strictly preventing lookahead bias**
- Observability & evaluability** — persistent execution trace, gate output, tool calls, citations for replay audit; **ConsistencyRunner** and **Arena** measure run-to-run stability

Engineering

- FastAPI + React + Pydantic Agent workbench — SSE streaming / run records / reconnection / replay; Reflection Checkpoint catches cross-gate contradictions; Arena 3-phase (**Decide → Vote → Tally**) blind multi-LLM peer review

Investment-Research Agent · Shinkai (in progress)

Project: github.com/Rain1601/shinkai

Active-discovery Agent for under-covered potential opportunities — **discovery → screening → hypothesis → evidence → deep research**; complements Uteki to form a **Research → Discovery** closed loop.

- Theme-driven supply-chain traversal (Mode B)** — given an industry theme (AI compute supply chain first-run completed), traces dependencies layer by layer from leaders (NVIDIA → wafer → optics → cooling), producing ranked candidates per layer
- Hypothesis-first + Research Graph** — generate hypothesis, then reverse-search evidence to validate / falsify; persists as a graph consumable by downstream agents (**Entity / Claim / Evidence / Question / Thesis** node types)
- Multi-persona critique + bounded self-iteration** — Planner / Reviewer / Optimizer + L2 critic counter-evidence search; hypothesis state machine drives iteration; hard tool-call budget prevents runaway loops

BACKEND PROJECTS

Apsara Conference Core System · Alibaba Cloud Intelligence

- Content moderation LLM-ization** (early-stage Agent landing in a traditional-business setting): LLM + rule-engine pipeline at **8M-file** scale, **95% auto-coverage, +70% reviewer productivity**
- Owned Apsara Conference core system (back-office / ticketing / exhibitor / online summit), supporting **87K+ registrations**, peak 200 QPM, **99.9% availability**
- 2024 / 2025 Apsara Conference **application development & technical PM**, coordinating multi-team delivery and change management

LONG-TERM DIRECTION

*Committed to long-term exploration of applying Agents to real-world complex problems — turning **excellent decision-making cognition and experience** into **continuously evolving systems**. My focus is the intersection of AI and investment research: structuring great investors' thinking frameworks into **engineered systems**, continuously iterated through Agents.*